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# Cretaceous paleomagnetism of the Fergana Basin and adjacent ranges, central Asia: tectonic implications

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ABSTRACT

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Paleomagnetic results have been obtained from Cretaceous red beds sampled at fifteen localities in the Fergana basin and adjacent ranges of the Tien Shan (Central Asia). The natural remanent magnetization of these rocks was accounted for by an Late Cretaceous overprint of normal polarity at fourteen localities and dual polarity, presumably primary component, at seven localities. No definite correlation was found between the declinations of the two components and structural trends; thus implying no oroclinal bending of the arcuate structures in the area studied. Inclination data are difficult to interpret due to probable tectonic movements, inclination errors of various types, and ambiguities in the Cretaceous reference data. All locality mean declinations were found to be deflected westward with respect to the Eurasian paleomeridian. The proposed counter-clockwise rotation of the Fergana block during the late Cenozoic is shown to account for the main features of the Alpine tectonics. In particular, the 200 km offset along the Talas-Fergana fault is connected with rotation, which contradicts the common idea that this displacement is of Late Paleozoic age. The horizontal movements are most probably not older than 10 Ma.

## Introduction

The Tien Shan tectonic realm is an example of an ancient mobile zone re-activated by younger tectonics. After Late Paleozoic deformation, almost no tectonic activity took place here during the Mesozoic and Paleogene. In the Neogene, both the Caledonian and Variscan belts were gradually remobilized and the most intense deformation was in the Pliocene–Quaternary. This activity is usually linked with the India–Eurasia collision (Molnar and Tapponnier, 1975). However, the relative intensity of the Variscan and Alpine tectonics is not clear. The prevailing opinion is that the Alpine movements inherited the older structural pattern and slightly reworked it. In contrast, other geologists think that the modern structural pattern is mainly Late Cenozoic and a result of Alpine movements (e.g., Yablonskaya, 1989).

Paleomagnetism may solve this problem, but the results reported so far are not conclusive. Data on the Cretaceous from various authors (Khramov, 1984), are contradictory, in terms of both sign and magnitude of rotation. The main body of the results is based on natural remanent magnetization (NRM) data or NRM after limited cleaning and does not satisfy modern reliability criteria. A result from Lower Cretaceous basalts from the north of the Fergana Basin (Eroshkin, 1991) was obtained after complete stepwise demagnetization and confirmed with the reversal test. However, its value is limited because only three lava flows were studied. Some data on Pliocene sediments from the Fergana Basin (Khaydarov, 1987) were interpreted as showing about 10° of counter-clockwise rotation. This explanation is not unique, however, since the declination anomaly might stem from primary tilts of sedimentary rocks which have accumulated in

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intermontane basins (Khaydarov, 1987). Recent deformation is also known west of the Tien Shan, in the Kyzyl-Kum Desert. Cenozoic paleomagnetic results have been reported from this area, but the main part of the collection was subjected to time cleaning only (Abdullaev et al., 1981). Thus, geodynamic models for Central Asia, where rotations have been hypothesized (Ulomov, 1974; Cobbold and Davy, 1988) have not been testable to date.

Cretaceous rocks were sampled in the Fergana Basin and adjacent ranges of the Southern Tien Shan in order to study Cenozoic horizontal movements. In total, about 400 oriented samples were taken from thirteen localities (Fig. 1). In addition, collections from three localities (9, 10 and, in part, 11) which had not been completely demagnetized (Bazhenov and Burtman, 1986) were restudied and incorporated into this paper.

## Geological setting and sampling

The Tien Shan comprises several Caledonian and Variscan fold belts. The post-Paleozoic history of this area differs to the north and south of the Talas-Fergana Fault (TFF) (Fig. 1). Apart from a few patches with Jurassic terrigenous coal-bearing sediments, there are no Mesozoic and early Cenozoic rocks to the north and northeast of the TFF. The succession here starts with terrigenous Eocene–Oligocene red beds with some basaltic flows near its base, overlain by a thick pile of Neogene sediments. In contrast, the succession to the south and southwest of the TFF is more complete. Grey sandstones and siltstones, often coal-bearing, of Early–Middle Jurassic age are overlain by variably colored Upper Jurassic sediments, Early Cretaceous terrigenous red beds of various grain sizes and Late Cretaceous rocks, often of marine origin. Above these there are Paleogene sediments of various origins and a thick pile of Neogene conglomerates and sandstones. Volcanic rocks are known just from a limited area in the north of the Fergana Basin, where a few basaltic flows are found in Lower Cretaceous rocks. Marine sediments (Upper Cretaceous and Eocene) are subdivided into stages and substages, whereas continental deposits (Lower Cretaceous and Neogene) are dated with much less accuracy. There are no pre-Neogene unconformities within this sequence, with the exception of some weak ones locally recognized near the Jurassic–Cretaceous boundary. Deformation started in the Miocene or slightly earlier and the main folding took place in the Pliocene.

The Talas-Fergana Fault is probably the most interesting structure in the Tien Shan. Various views on its evolution have been expressed (for a review see Burtman (1964); Tapponnier and Molnar (1979)). Most geologists are now of the opinion that it is a dextral strike-slip fault, but the age of movement is uncertain. The only well established fact is that the movement is post-Early Permian in age, and most of it is thought to be Late Permian or Early Triassic in age (Burtman, 1964). However, this conclusion is based on an analysis of Paleozoic formations and the upper age limit of the movement has not been determined directly. The highest amount of right-lateral offset (about 200 km) is found in the central part of the TFF. At both terminations the offset is either much less or absent. There are large vertical displacements without noticeable strike-slip movement at the northern termination; whereas at the southern one, west of the China-Kirghizia border, the fault branches and disappears before reaching the Tarim platform.

A Quaternary, dextral, strike-slip displacement along the TFF of about 10 km is also well established (Rantsman and Pshenin, 1963; Wallace, 1976). The recent movements were greatest in the central part of the fault and nearly disappear at its terminations, to the northwest of the Chatkal Range and near the Tarim platform. Thus the spatial distributions of the total displacement along the TFF and its Quaternary part are very similar. The velocity of recent movements ranges from 0.25 to 1.5 cm/yr and is about 1 cm/yr on average (Trifonov et al., 1990). A hypothetical post-Cretaceous displacement of about 60 km (Verzilin, 1968) is based on the only small outcrop of red beds north of the TFF, where the age of the rock is uncertain.

Lower Cretaceous red beds were chosen for sampling because they are widespread in the Fergana Basin and adjacent ranges, often well ex-

posed and suitable for paleomagnetic study. Red beds accumulated under conditions where noticeable primary tilts are unlikely. Lower Cretaceous rocks are mainly red, medium to fine grained sandstones, although only coarse-grained sandstones were studied at locality 1; grey varieties were sampled in the southern part of the area too. As a rule, mudstones and siltstones are poorly exposed and often very friable and they were well sampled only at localities 3 and 15. Cretaceous

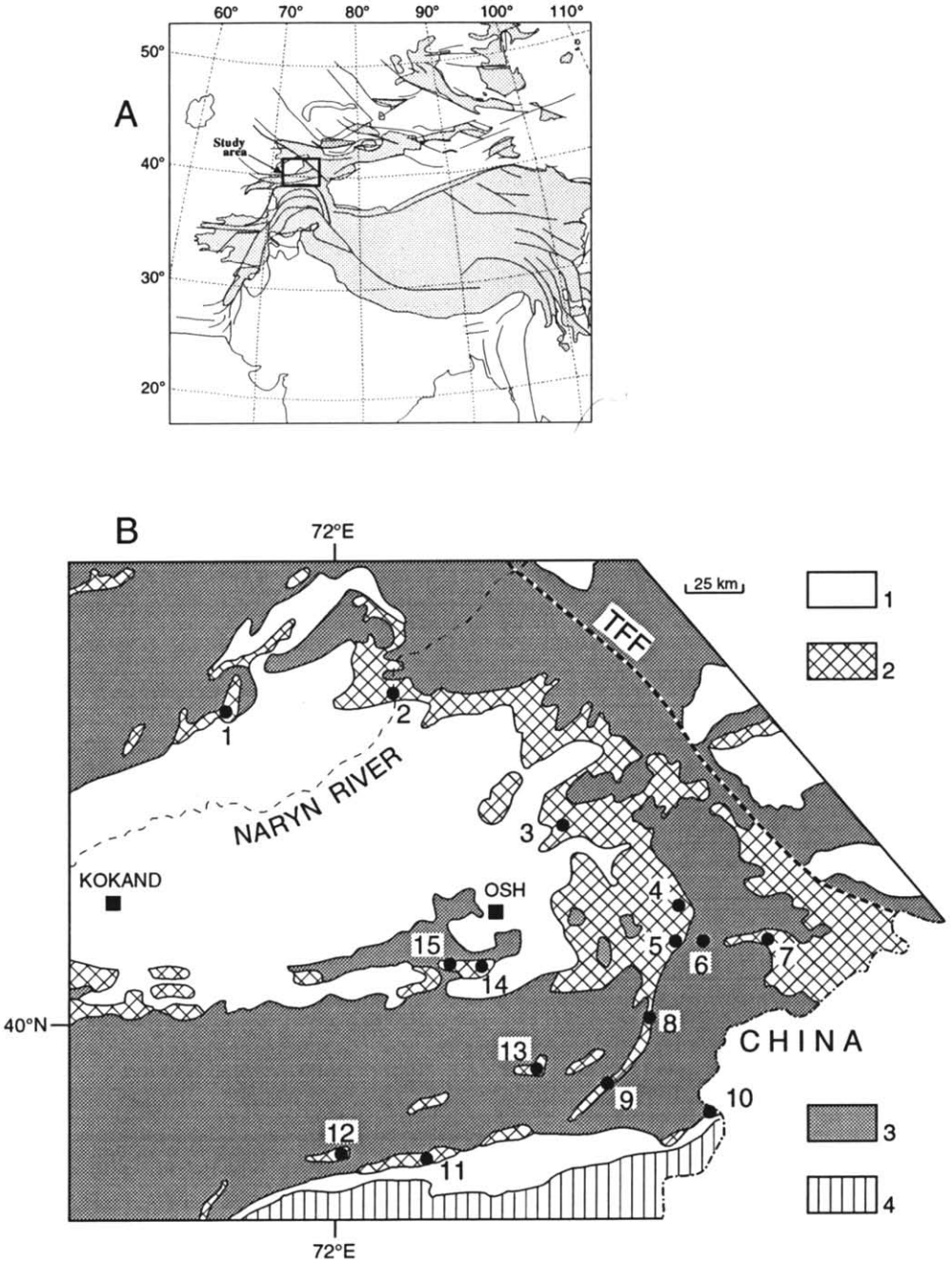



Fig. 1. (a) Cenozoic deformation in Central Asia (after Cobbold and Davy, 1988). Map shows traces of major faults and areas with elevation greater than 2 km (stippled area). (b) Schematic geological map of the Fergana basin and adjacent ranges and positions of sampling localities. *1* = Neogene and Quaternary rocks; *2* = Paleogene and Mesozoic rocks; *3* = Paleozoic rocks; *4* = the Pamirs; *TFF* = the Talas-Fergana fault.

rocks are often limited to narrow fault-bounded folds and graben-like structures and various attitudes could be studied at nine localities (further on, a collection from a fold limb is referred to as a section). At locality 6, sampling was carried out on a thrust sheet "squeezed" between Paleozoic rocks. Jurassic grey sandstones were also studied at localities 2, 12 and 14.

The true thickness studied at a locality varied from several tens to several hundreds of meters. One hand sample oriented with magnetic compass was taken from a layer, sampling points being spaced several meters apart along a cross-section. A total of 20–40 samples were taken from each locality. In addition, 16 pebbles of red beds were collected from a layer of intra-formational conglomerate at locality 14.

## Treatment

Two to four cubic specimens were cut from each hand sample in the field using a portable diamond saw. Two specimens from each hand sample, including those from pebbles, were thermally demagnetized in 10–12 steps. All measurements were made with a Czechoslovakian spinner magnetometer JR-4 with the noise level of 0.05 nT. The results were analyzed with stereonets and orthogonal diagrams. Components of the NRM isolated from two specimens of a hand sample, were used for calculation of sample means, which, in turn, were used for calculation of section means and locality means. Converging remagnetization circles (Halls, 1976) were used on the specimen level. Since each locality, or each section at a locality, was a relatively long monoclinal section, the fold test was performed as suggested by McFadden and Jones (1981). A similar approach, based on $F$ statistics, was used to compare any mean directions.

## Results

NRM directions in situ are scattered and mostly point downward. A few samples from the southern part of the area (localities 9, 10, 12) were found to be remagnetized by the present day field (PDF) but, in most samples, no recent component was identified above 300°C. Coarse-grained sandstones from locality 1 show a chaotic scatter of specimen directions and were discarded. Jurassic grey sandstones sampled at localities 2, 12 and 14, and grey-colored Lower Cretaceous rocks yielded no consistent results after removal of unstable components aligned along the PDF. Thus all the results are from red beds.

After removal of the low-temperature remanence, a component of normal polarity, represented by a linear trajectory above 200° or 300°, was isolated in most samples from 14 localities (Fig. 2). The unblocking temperature spectra of this component is more-or-less uniform and its intensity is usually above 50% of the NRM. In some samples, this component persisted up to 650° and was the only component found (Fig. 2a,b). However, orthogonal plots often either bypass or pass over the origin (Fig. 2c,d), or another linear trajectory can be recognized (Fig. 3), thus implying that another component is present. Some samples revealed an apparently uni-vectorial trajectory and another component could be suspected only because of the steady shift of the vector end-point on the stereonet (Fig. 2e–h).

The linear segments of orthogonal plots corresponding to another component are usually found above 550°C. The isolation of the second component is easier if it has reversed polarity (Fig. 3a,b); even in this case, the reversed component often cannot be adequately resolved because its unblocking temperature spectrum is very narrow (Fig. 2d). If both components have normal polarity their separation is often difficult because of the close agreement of component directions (Fig. 3).

Analysis of demagnetization data (Figs. 2–4) shows that unblocking temperature spectra of the NRM components vary from sample to sample and often overlap. Hence it is impossible to determine precise "temperature windows" for the component identification. The following procedure was adopted. Irrespective of whether the orthogonal plot decayed to the origin or bypassed it, all linear segments above 200° or 300° were treated as one component ($A$), unless another linear segment could be recognized. The component $A$ always has normal polarity and its direc-

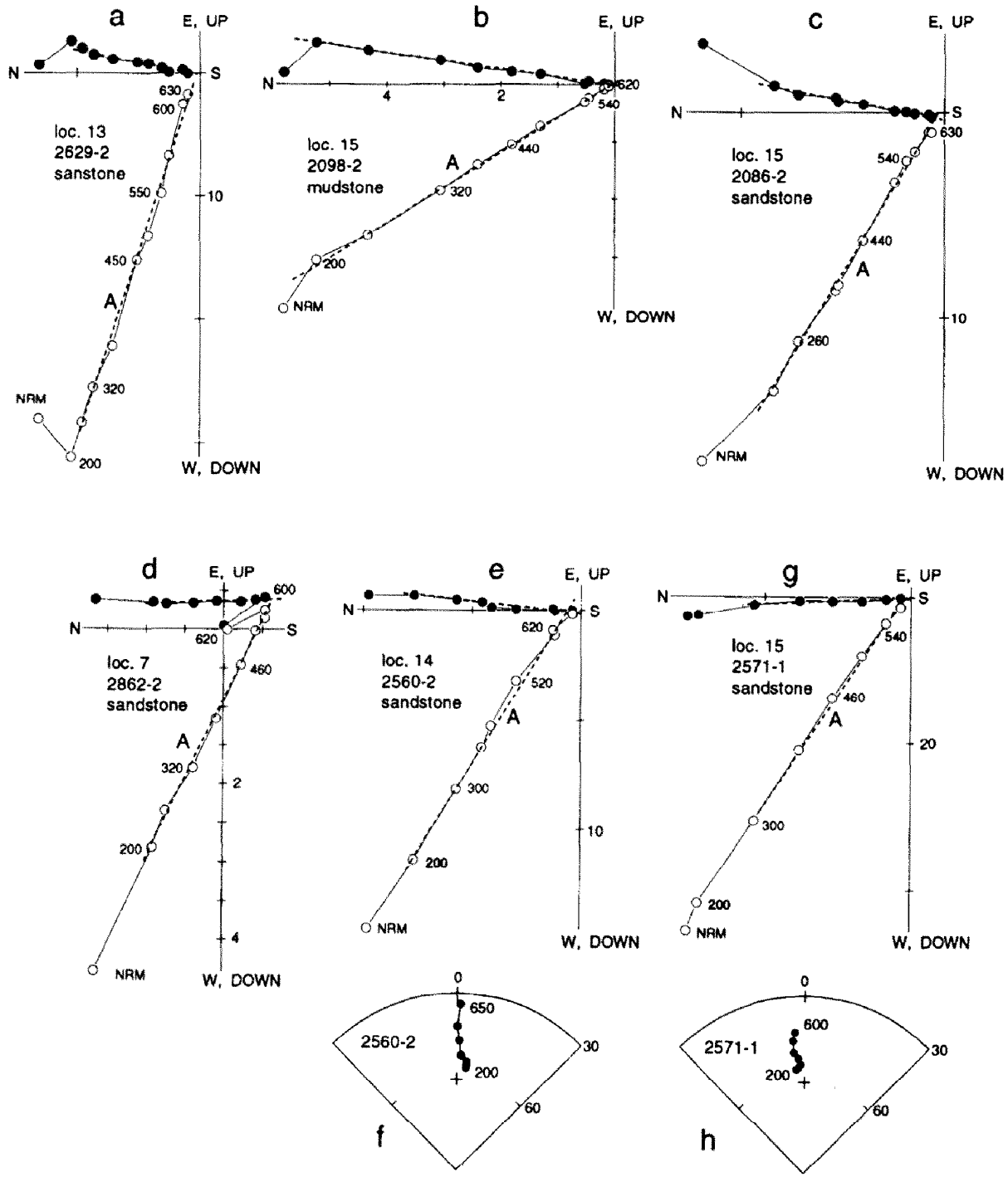



Fig. 2. Representative orthogonal plots for samples where only the component $A$ was isolated. (a) and (b) Univectorial magnetization. (c) and (d) Orthogonal plots slightly (c) or clearly (d) bypassing the origin. (e) and (g) Examples of the apparently rectilinear decay to the origin. (f) and (h) Equal area projections of vector end points for the same samples. Dashed lines = isolated component $A$; dots = vector end points projected onto the horizontal plane; circles = vector end points projected onto the vertical plane. Data are plotted in stratigraphic coordinates; magnetization units on axes are in $10^{-3}$ A/m; temperatures are in degrees Celsius.

tions proved to be tightly clustered (Table 1). All the other components recognized at high temperatures were also treated as one component ($B$). Although more scattered than the component $A$, component $B$ is also consistent within a locality (Table 2). It is reversed at localities 3 and 4 and predominantly reversed at locality 7. In contrast, at localities 6, 14 and 15, this component is normal and it is predominantly normal at locality 5. The reason for this is that different parts of the Cretaceous sequence were studied at adjacent localities 4–7 (Fig. 1), which accounts for the differences in the polarity of component $B$; still other parts of the sequence were most probably studied at localities 14 and 15.

No changes of NRM intensity which could

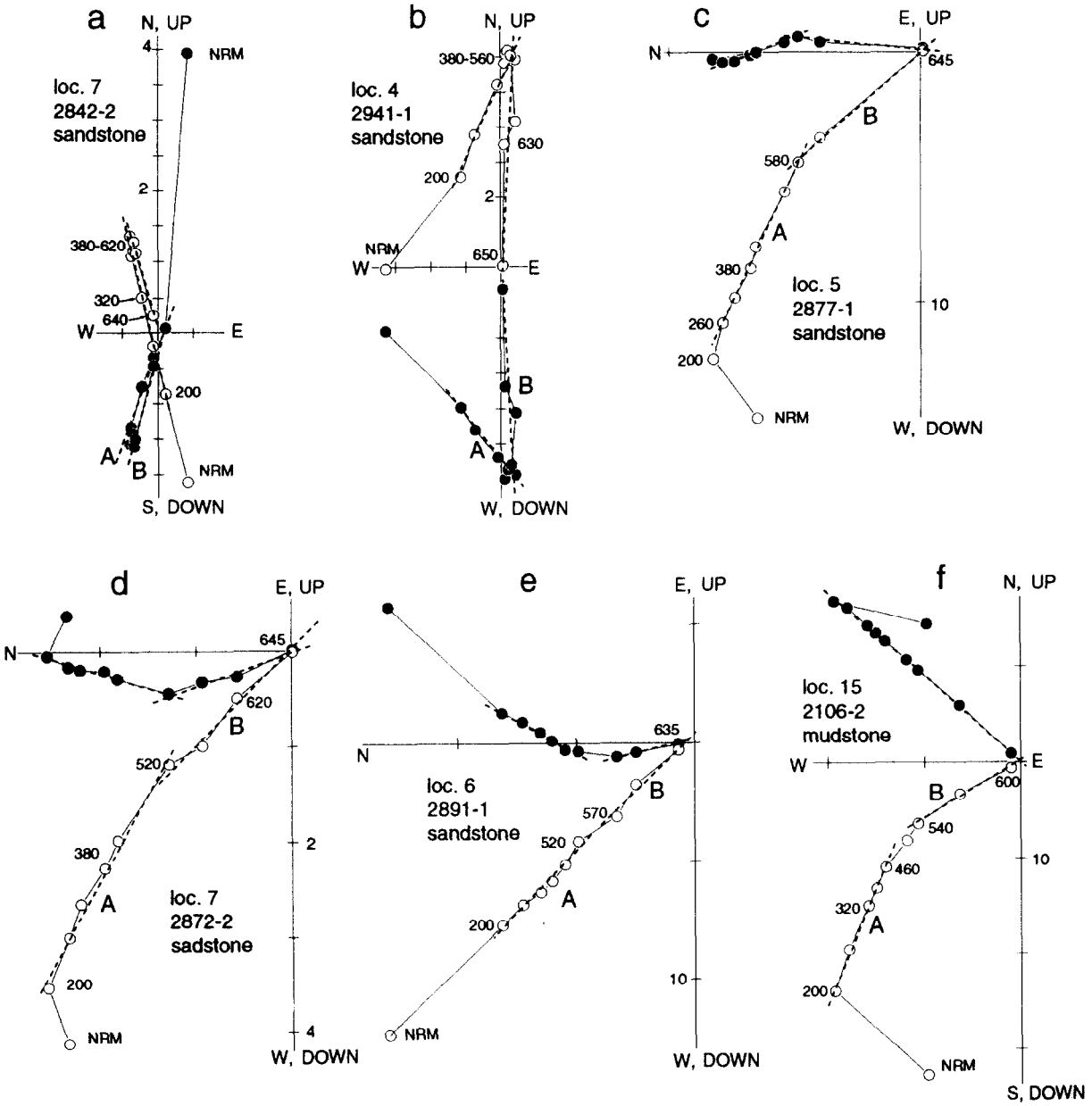



Fig. 3. Representative orthogonal plots for samples where the components $A$ and $B$ are isolated. (a) and (b) The component $B$ has reversed polarity. (a) locality 7; (b) locality 4. (c)–(f) The component has normal polarity and differs from the component $A$ both in declination and inclination (c and d), only in declination (e) and only in inclination (f). Other notation as in Fig. 2.

TABLE 1

Overprint directions (component *A*) of Late Cretaceous age

| Location | In situ | | | | | Tilt corrected | | | |
|---|---|---|---|---|---|---|---|---|---|
| | *N* | *D* (°) | *I* (°) | *K* | $\alpha_{95}$ (°) | *D* (°) | *I* (°) | *K* | $\alpha_{95}$ (°) |
| 2 | 25/18 | 352 | 20 | 4 | 16.5 | 353 | 54 | 21 | 7.2 |
| 3 | 23/15 | 0 | 38 | 56 | 4.9 | 359 | 51 | 57 | 4.8 |
| 4 | 20/11 | 27 | 33 | 126 | 3.8 | 333 | 58 | 74 | 4.9 |
| 5 | 22/19 | 327 | 26 | 29 | 6.0 | 358 | 56 | 32 | 5.8 |
| 6 | 15/13 | 351 | −3 | 57 | 5.1 | 344 | 60 | 52 | 5.4 |
| 7 | 42/32 | 13 | 60 | 5 | 11.5 | 355 | 54 | 24 | 5.0 |
| 8 | 33/20 | 41 | 23 | 4 | 14.8 | 347 | 56 | 33 | 5.5 |
| 9 | 35/18 | 12 | 24 | 4 | 18.7 | 9 | 47 | 32 | 5.8 |
| 10 | 32/17 | 116 | 47 | 31 | 6.3 | 14 | 49 | 40 | 5.4 |
| 11 | 35/18 | 350 | 38 | 10 | 10.6 | 0 | 59 | 53 | 4.6 |
| 12 | 26/13 | 3 | 5 | 17 | 9.3 | 23 | 56 | 19 | 8.9 |
| 13 | 26/17 | 47 | 54 | 39 | 5.4 | 356 | 60 | 62 | 4.3 |
| 14 | 20/13 | 351 | 15 | 61 | 5.0 | 341 | 62 | 60 | 5.0 |
| 14 * | 16/16 | – | – | – | – | 345 | 66 | 38 | 5.7 |
| 15 | 43/42 | 26 | 63 | 9 | 7.1 | 6 | 53 | 40 | 3.4 |
| *Sandstones* | | | | | | | | | |
| 15 | 26 | 25 | 65 | 6 | 9.8 | 7 | 57 | 82 | 3.0 |
| *Mudstones* | | | | | | | | | |
| 15 | 16 | 26 | 60 | 12 | 10.1 | 5 | 45 | 32 | 6.2 |

* Pebbles

Localities numbered as in Fig. 1. *N* = the number of samples studied/accepted; *D* = mean declination; *I* = mean inclination; *K* = Fischer precision parameter; $\alpha_{95}$ = radius of confidence circle.

have been related to magnetite, were found in demagnetization data (Fig. 4). Rock magnetic studies show that haematite is the only NRM carrier in red beds, even in the samples which are completely demagnetized at 600°. Thus, all the NRM components reside in haematite. A large amount of magnetite was found only in grey sandstones, which yielded no consistent data.

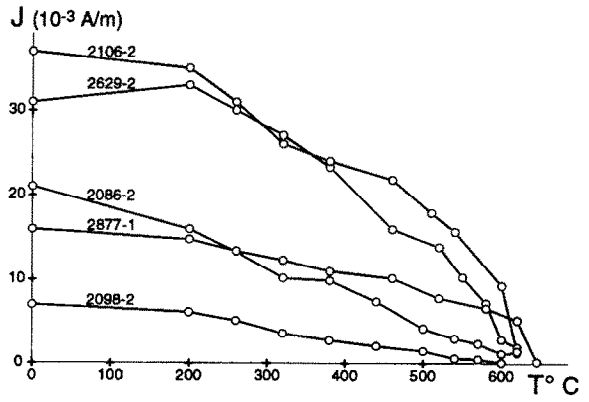



Fig. 4. Representative plots of NRM intensity versus temperature.

TABLE 2

Mean directions of component *B*

| Location | In situ | | | | | Tilt corrected | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | *N* | *D* (°) | *I* (°) | *K* | $\alpha_{95}$ (°) | *D* (°) | *I* (°) | *K* | $\alpha_{95}$ (°) | *P* |
| 3 | 23/9 | 355 | 21 | 40 | 7.4 | 354 | 35 | 39 | 7.4 | R |
| 4 | 20/13 | 21 | 7 | 26 | 7.6 | 0 | 40 | 29 | 7.3 | R |
| 5 | 22/8 | 331 | 10 | 16 | 12.3 | 347 | 41 | 20 | 11.1 | Nr |
| 6 | 13/7 | 356 | −12 | 24 | 10.9 | 357 | 51 | 26 | 10.5 | N |
| 7 | 42/17 | 17 | 45 | 5 | 15.5 | 5 | 44 | 32 | 6.0 | Rn |
| 14 | 20/6 | 341 | 1 | 38 | 9.3 | 333 | 46 | 28 | 10.7 | N |
| 15 | 46/14 | 25 | 69 | 11 | 11.5 | 355 | 49 | 48 | 5.4 | N |

*P* = polarity; R = reversed; Rn = predominantly reversed; N = normal; Nr = predominantly normal.

The other notation is the same as in Table 1.

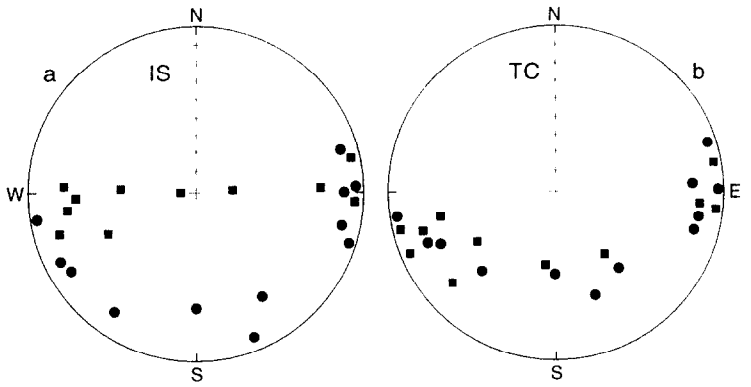



Fig. 5. Distribution of normals to remagnetization circles for two limbs of the fold (locality 7) (a) before and (b) after tilt correction. Dots and squares denote the data from different limbs. All vectors are projected onto the lower hemisphere.

The end-points of many specimens shifted along great circles during cleaning. For each locality, normals to unit remagnetization circles are restricted to long narrow bands. Where two or more fold limbs were studied at a locality these bands from each monoclinal section overlap after tilt correction, but diverge in situ (Fig. 5). The direction of the least-dispersed component, determined as the normal to the best-fitting plane (Halls, 1976), is in agreement with the component *A* mean vector at each locality; the angular difference between them varies from 1° to 11°, about 6° on average, with the exception of locality 12, where this difference is about 20° (Tables 1 and 3). Here, several samples were found to be completely remagnetized by the PDF and the distribution of sample mean vectors is somewhat smeared towards the same direction. It may be that a recent component was not completely removed here; so, the result obtained by Halls (1976) have been used for the interpretation (Table 3).

A well-clustered component was isolated from pebbles from an intra-formational conglomerate at locality 14 (Fig. 6). Directions of this component derived either from remagnetization circles or orthogonal plots, are very close to the component *A* for this locality (Tables 1 and 3). Paleomagnetic vectors of some samples swung away from the cluster at 600° or 620° (Fig. 6), whereas two specimen directions remained consistent; thus these swings are not due to a laboratory-induced components. Unfortunately, these samples were completely demagnetized at the next step and another component was not isolated.

Mudstones were mostly sampled at locality 3, whereas sandstones strongly prevailed at the other localities. Only at locality 15 was it possible to

TABLE 3

Directions of the least-dispersed component derived from converging remagnetization circles

| Location | *n* | *D* (°) | *I* (°) | $O_1$ (°) | $O_2$ (°) | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|
| 3 | 17/17 | 354 | 54 | 10.9 | 4.8 | 7.2 |
| 4 | 23/23 | 347 | 54 | 9.2 | 5.2 | 6.9 |
| 5 | 24/22 | 359 | 48 | 8.2 | 5.6 | 6.8 |
| 6 | 18/16 | 341 | 50 | 10.7 | 6.5 | 8.3 |
| 7 | 51/47 | 356 | 50 | 5.7 | 4.2 | 4.9 |
| 8 | 31/22 | 355 | 56 | 7.9 | 5.2 | 6.4 |
| 9 | 17/14 | 18 | 50 | 12.7 | 6.7 | 8.4 |
| 10 | 17/16 | 13 | 50 | 17.7 | 9.4 | 11.8 |
| 11 | 14/11 | 357 | 57 | 9.8 | 4.0 | 6.2 |
| 12 | 12/10 | 352 | 50 | 13.6 | 9.9 | 11.7 |
| 13 | 19/17 | 354 | 65 | 9.0 | 7.2 | 8.1 |
| 14 | 20/20 | 338 | 70 | 10.7 | 4.3 | 6.9 |
| 14 * | 9/9 | 343 | 71 | 17.5 | 12.9 | 15.0 |
| 15 | 36/34 | 359 | 49 | 5.1 | 3.6 | 4.1 |

* Pebbles

*n* = number of remagnetization circles found/accepted; *D* and *I* = declination and inclination of the least-dispersed component in stratigraphic coordinates; $O_1$ and $O_2$ = the semi-axes of confidence oval; $\alpha_{95}$ = radius of confidence circle calculated as the square root of ($O_1$ $O_2$).

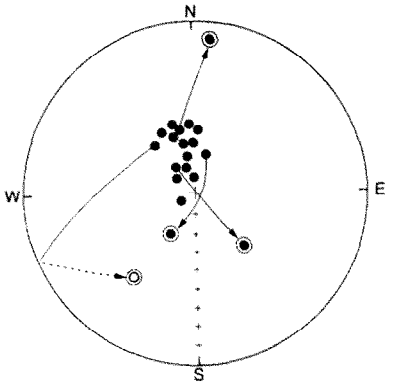



Fig. 6. Paleomagnetic data from pebbles. Tilt-corrected directions of the component *A* are shown by dots. Encircled dots denote paleomagnetic vectors obtained at the last heating step. Dots and solid lines are projected onto the lower hemisphere of equal area projection circles; circles and dashed lines are projected onto the upper hemisphere of equal area projection.

compare data from different rock types (Fig. 2b,c). Mean declinations of the component *A*, from sandstones on the one hand and mudstones and marls on the other, are very similar, but mean inclinations differ significantly (Table 1). Component *B* was isolated from a limited number of samples at this locality and it was not possible to compare its inclination in various rock types. Apart from this difference in inclination, no other correlation was found between lithology, on the one hand, and unblocking temperatures for components or component directions on the other (Figs. 2 and 3).

Results were obtained only from one fold limb at localities 10–12. As a result, of the nine localities where two of more fold limbs were sampled, the fold test (McFadden and Jones, 1981) could only be performed for 6 localities, for each component separately (Table 4). The test is positive for component *A* at localities 2, 9 and 11 and component *B* at localities 7 and 15, whereas the calculated *F* statistics slightly exceed the critical values for component *A* at localities 7, 8 and 15 (Table 4). For localities 7 and 8, it is most probably due to the fact that one of the section means at each locality is based on a limited number of samples and the proper averaging of paleomagnetic noise is not achieved. When the result from the smallest section is omitted, the fold test becomes positive, as illustrated for locality 7 (Table 4). All section means are well-defined at locality 15, but the percentage of sandstones and mudstones within each section is different. When performed for each rock type separately, the fold test is positive. Thus, components *A* and *B* predate the folding at these six localities.

Paleomagnetic data are from monoclines at the other eight localities. Everywhere, the scatter of bedding poles is considerably less than that for component directions in any coordinates, and hence precision parameters are similar, with the exception of locality 4 (Table 1). Here, the dispersion of bedding poles is very small ($K = 160$). Bed surfaces could be slightly uneven, for instance, due to syn-sedimentary undulations, thus inducing small errors in bedding measurements. These errors will not affect the in situ data, but will increase the dispersion in stratigraphic coordinates. Therefore, the precision parameter ratio cannot be used here for magnetization dating.

Locality means for both components are much better grouped after tilt correction than in situ (Fig. 7). However, the calculated *F* statistics for

TABLE 4

Results of the fold test application

| Location | *C* | *N* | *G* | *DF* | *IS* | *TC* | *CV* |
|---|---|---|---|---|---|---|---|
| 2 | A | 18 | 3 | 4/30 | 32.2 | 0.4 | 2.7 |
| 7 | A | 32 | 4 | 6/56 | 40.0 | 3.9 | 2.3 |
| 7 | A * | 28 | 3 | 4/50 | 45.0 | 1.9 | 2.6 |
| 7 | B | 15 | 3 | 4/24 | 32.5 | 2.2 | 2.8 |
| 8 | A | 18 | 3 | 4/30 | 59.6 | 3.7 | 2.7 |
| 9 | A | 18 | 2 | 2/32 | 117 | 2.8 | 3.3 |
| 11 | A | 16 | 3 | 4/26 | 31.4 | 1.5 | 2.7 |
| 15 | A | 42 | 3 | 4/78 | 76.2 | 3.7 | 2.5 |
| 15 | B | 14 | 3 | 4/22 | 24.6 | 1.6 | 2.8 |
| [6] | A | 148 | 6 | 10/286 | 18.4 | 3.6 | 1.9 |
| All | A | 265 | 14 | 26/502 | 22.3 | 5.9 | 1.5 |
| All | B | 74 | 7 | 12/134 | 17.4 | 3.4 | 1.8 |

* The result from the least studied section was omitted (see text).

[6] = the test for mean directions from the above-listed six localities; All = the test for all localities; *C* = the NRM component; *N* = number of unit vectors used; *G* = the number of mean vectors (sections); *DF* = the number of degrees of freedom; *IS* and *TC* = computed *F* statistics before and after tilt correction, respectively; *CV* = the critical value at the 95% confidence level.

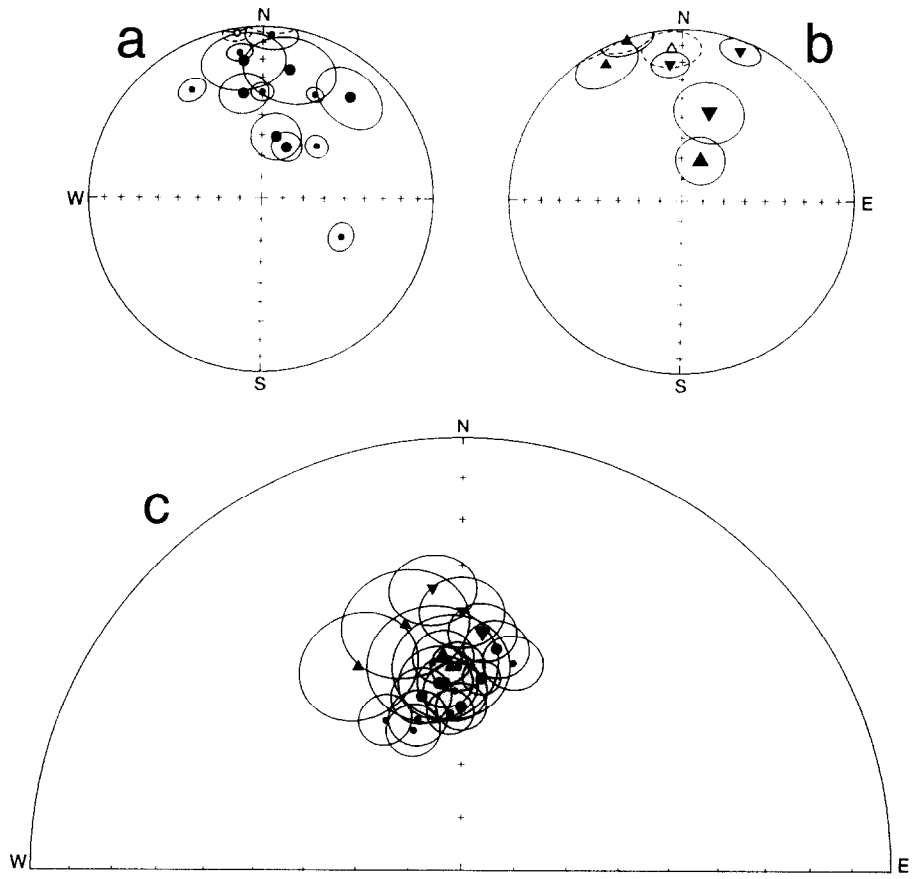



Fig. 7. Stereoplots of mean directions of (a) component *A* and (b) component *B*, in situ. (c) Both components after tilt correction. Dots = the component *A* data; triangles = results from localities where the component *B* has normal or predominantly normal polarity; inverted triangles = results from localities where the component *B* has reversed or predominantly reversed polarity. All the data for component *B* are shown as normal polarity directions. Solid symbols = projected onto the lower hemisphere; outlined symbols = projected onto the upper hemisphere; large symbols = data from localities where the fold test was performed; small symbols = data from localities where the fold test was not performed.

two sets of mean directions are larger than the corresponding critical values (Table 4). Even if the results from the six localities where the fold test was performed are taken for component *A*, the tilt-corrected locality means are statistically different. Thus the data set for any two components cannot be considered a random sample from the same general population.

## Dating of the NRM components

Component *A* was often the only component identified above 300°C, but it is most probably a younger overprint, judging by the orthogonal plots (often not passing through the origin) and the negative conglomerate test. Since the data came from red beds and the magnetization resides in haematite, component *A* is likely to be chemical in origin (CRM). Its pre-folding age was proven for the data from six localities (Table 3). Demagnetization characteristics, the consistency of the results derived from orthogonal plots and those from converging remagnetization circles, and, most important, the great improvement in clustering of the locality means after tilt correction, testify to the pre-folding age of component *A* everywhere. Mesozoic and Paleogene rocks in the Fergana Basin and adjacent ranges were mainly folded in the Pliocene. However, close to the Oligocene–Miocene boundary, conglomerates up to several kilometers thick started accumulating in the basin, implying differentiated range-and-

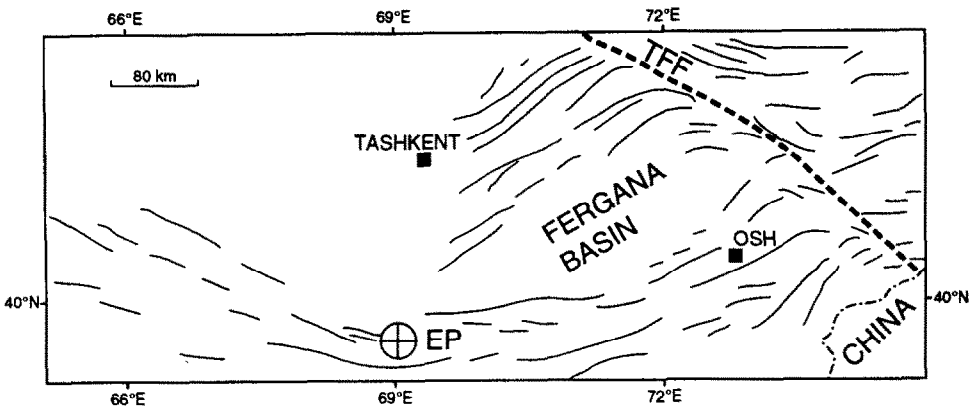



Fig. 8. The simplified scheme of axes of the Alpine structures (after Makarov et al., 1982). *TFF* = Talas-Fergana fault; *FB* = Fergana Basin; *EP* = the position of the Euler pole.

basin relief and, therefore, some tectonic activity. Thus, component *A* is pre-Miocene in age.

Polarity of this remanence is normal everywhere. Moreover, similar Lower Cretaceous red beds are common in the southwestern part of the Tien Shan and in Tadjik Basin, where a similar normal polarity overprint was also found (Pozzi and Feinberg, 1991). Therefore, this overprint is a regional phenomenon. A long, normal polarity, interval lasted from the early Aptian up to the Campanian, with frequent polarity changes both before and after this interval. Partial remagnetization of Lower Cretaceous red beds is much more likely to have taken place during the long normal interval than during a much shorter earlier or later chron.

The dependence of inclination on lithology found at locality 15 is not the inclination error, which is common for detrital remanent magnetization, since component *A* is not primary. Cretaceous rocks at this locality outcrop on two limbs of a syncline, with dips not exceeding 40°, and show no signs of internal strain such as cleavage. This means that the difference in inclinations cannot be attributed to internal deformation. The only explanation I could find is that the difference in inclination is due to compaction: the steeper vectors are from sandstones which are not liable to strong compaction, whereas the shallower ones are from more easily compactable mudstones and marls. The compaction of sediments takes place mainly during the early stages of diagenesis, at burial depths of a few hundred meters (Meade, 1966). Such a thickness of overlying sediments had certainly accumulated by the end of the Cretaceous. Thus, two lines of evidence point to a Late Cretaceous (but pre-Campanian) age for component *A*.

The fold test points to pre-folding age for component *B*. It is reasonable to assume that, being more thermally stable than component *A*, it is older and primary. The presence of both polarities also favors an Early Cretaceous age.

## Tectonic implications

The main trends of the Alpine structures vary throughout the area studied (Fig. 8, after Makarov et al. (1982)) and approximately follow the Paleozoic basement. If mean declinations are plotted against the strikes of Alpine structures, an angle between the regression line and *X* axis is related to the origin of strike variations (Bazhenov, 1982). Also, a significant data scatter with respect to the regression line implies that rotations were largely unrelated to strike variations, provided that the quality of data is good. The regression lines are nearly horizontal for both components (Fig. 9), but a certain scatter of data is present. Thus, the observed structural pattern is not the result of oroclinal bending, but some rotations could have taken place here. The result from the thrust sheet (locality 6) does not deviate significantly from the rest and, therefore, thrusting was not accompanied by rotation.

Large rotations deduced from structural data

near the Talas-Fergana Fault are thought to have taken place during the Late Paleozoic (Burtman, 1976). Unfortunately, paleomagnetic data on Upper Paleozoic rocks (Burtman and Gurary, 1973; Klishevich et al., 1987) are not based on adequate cleaning or confirmed by field tests. The Cretaceous data (localities 4–8) lead to the conclusion that if any large rotations occurred here, then they should be pre-Cretaceous in age.

At the next step, the results were compared to reference paleomagnetic data from the Eurasian plate. There are two versions of the Cretaceous apparent polar wander path (APWP) for Eurasia: the older path (e.g., Irving, 1977) and another one by Besse and Courtillot (1991), whose revised APWP is mainly based on the synthesis of the APWP's from several lithosphere plates. The Tien Shan results are compared with both, henceforth labelled E and BC, respectively. For the E pole, a mean pole at 73°N/154°E ($\alpha_{95} = 3.7°$) (Bazhenov and Shipunov, 1985) was chosen, as this pole is based on a larger number of unit poles and was obtained by intersection of Cretaceous paleomeridians; thus a possible bias due to inclination errors in sediments was taken into account. 90 Ma and 120 Ma poles (Besse and Courtillot, 1991) were used as the BC references for components *A* and *B*, respectively.

The nine mean inclinations for component *A* do not differ significantly from the predicted E value. Of the remaining five, shallowing of inclinations was proven for the result from locality 15. In contrast, some results are certainly steeper than the Late Cretaceous BC value (Fig. 10). The inclinations of component *B* and the result for Lower Cretaceous basalts (Eroshkin, 1991), are well below the E reference value and are in better agreement with the other one (Fig. 10). In

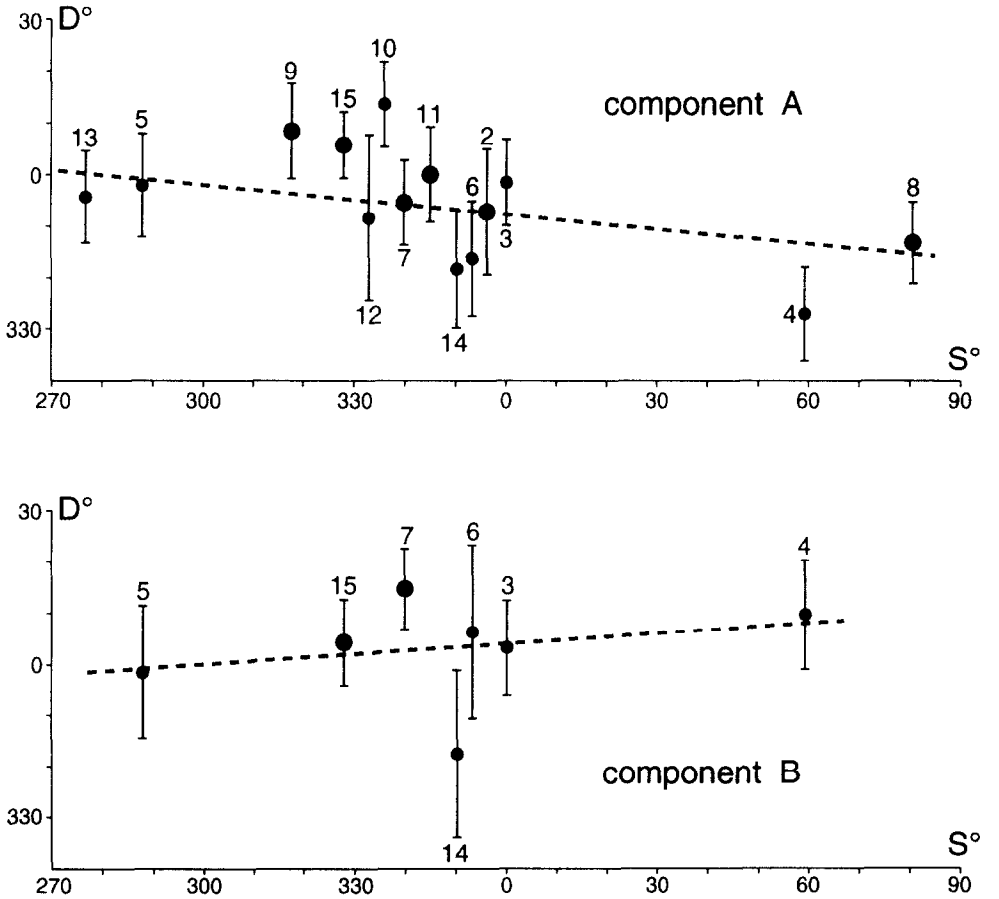



Fig. 9. The plots of locality mean declinations (*D*) together with the error bars versus strike (*S*) of Alpine structures. Large dots = the results confirmed by the fold test; small dots = results not confirmed; dashed line = the regression line.

general, the component *A* inclinations are in better agreement with the earlier versions of the APWP than with the revised one; the situation is reversed for component *B*. At the same time, both sets of inclinations seem to be slightly shallow with respect to the corresponding reference values. All hypotheses on the Cenozoic evolution of this region agree that tectonic transport was directed approximately northwards and not southwards. Unfortunately, it is not possible to separate the contributions of tectonics and inclination errors in sediments.

All versions of the Eurasian APWP predict a reference declination between 15° and 20° for the Fergana Basin. Moreover, similar declinations are also predicted by the Paleogene reference poles, and hence the interpretation of declination data is almost independent of the reference data used. Locality mean declinations are deflected westward with respect to the E reference direction and distributed fan-like over about 40° for component *A* and slightly less for component *B* (Fig. 11). On average, the confidence limits of mean declinations for these two components are 9° and 11.5°, respectively, which is much less than the fan angles. Thus, each data set cannot be considered a sample from the same general population. With respect to the reference declinations, the values of rotation, *R* (Demarest, 1983) are statistically insignificant for locality 10 and marginally

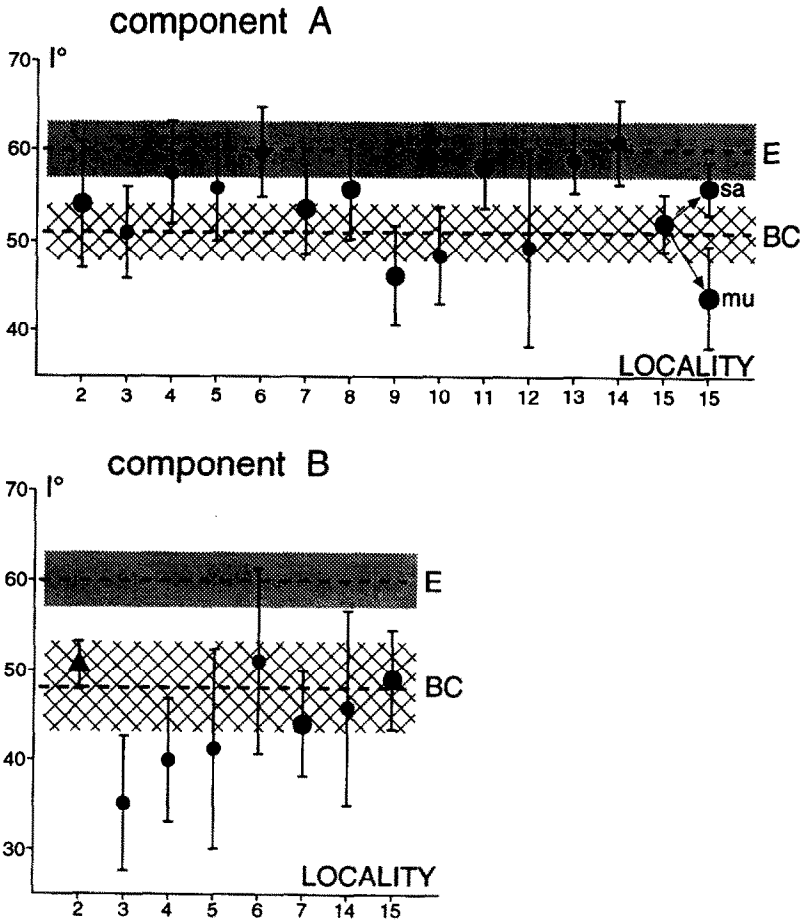



Fig. 10. The comparison of the measured and reference inclinations with their error bars shown as vertical lines and striped bands, respectively. Large dots = results confirmed by the fold test; small dots = unconfirmed results; triangle = a result from Lower Cretaceous basalts (Eroshkin, 1991); *sa* = data on sandstones, *mu* = data on mudstones from location 15; *E* and *BC* = the reference inclinations from Bazhenov and Shipunov (1985) and Besse and Courtillot (1991), respectively; 90 Ma and 120 Ma = BC poles were used for the component *A* and component *B* data, respectively.

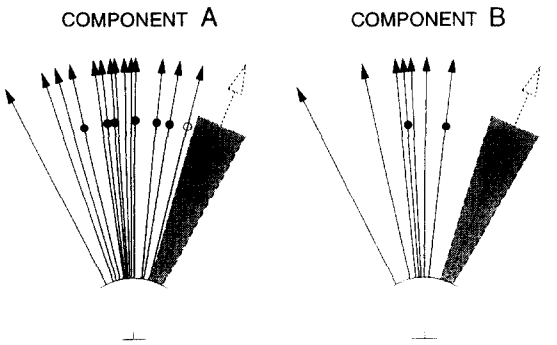



Fig. 11. Locality mean declinations together with the $E$ reference direction. Shaded area = confidence limits for the reference direction; arrow with circle = the result (location 10) which does not differ significantly from the $E$ direction; arrows with dots = localities where the fold test was performed.

significant for locality 9, from the southeast of the area. Otherwise, no correlation was found between the locality position and the angle of rotation.

It is logical to assume that the entire area studied (the Fergana block) was rotated counter-clockwise, but this regional movement might have been complicated by local movements, judging by the results of the regional test (Table 4) and the analysis of the plots of declination versus strike (Fig. 9). The mean $R$ value, 25°, is strictly valid only if local rotations are symmetrical with respect to the regional movement. The two types of rotation cannot be separated from paleomagnetic data alone and the mean $R$ value should be treated with some caution. One can conclude only that the Fergana block did not behave as a rigid body. Paleogene paleomagnetic data from three localities in the Fergana Basin also point to a similar counter-clockwise rotation (Thomas et al., 1992).

In the south Tien Shan, the strikes of the Alpine structures trend WNW–ESE west of meridian 69°E and mostly W–E east of this meridian (Fig. 8). The change in strikes (about 20–30°) is close to the mean $R$ value and oroclinal bending of this arcuate structure is hypothesized. If this is the case, the area to the east is rotated in accordance with paleomagnetic data. The Euler pole should be in the apex of the arc, since any other position would lead to the disruption of Paleozoic structures. As a result of the rotation, the whole area should be displaced northwards, especially its eastern part. It is logical to connect such movement with the impact of the Pamirs, as deduced from structural (Molnar and Tapponnier, 1975) and paleomagnetic (Bazhenov and Burtman, 1986) data.

There are no Cretaceous rocks to the north of the TFF, but Paleogene paleomagnetic results revealed no Cenozoic rotations in this area (Thomas et al., 1992). This fault is most probably the northeastern boundary of the rotated block. The TFF strike is oblique in relation to the proposed direction of movement, and both compression and strike-slip faulting took place here. This fault is now in an elevated area, often coinciding with ridge crests up to 5,000 m above sea level, and the uplifts are of the Late Cenozoic age. The compression associated with the rotation of the Fergana block accounts for this phenomena.

The total right-lateral offset of about 200 km along the TFF is post-Early Permian in age and is usually attributed to Late Permian and/or Early Triassic tectonics (Burtman, 1964). Quaternary movement, with an average velocity about 1 cm/yr, is well established along the TFF (Trifonov et al., 1990) and its distribution along the fault is similar to that for the total displacement. The 200 km offset could have occurred during the last 20 Ma, provided that the velocity was constant. Alpine deformation also started in the Tien Shan about 20 Ma ago (Early Miocene). The distance between the proposed position of the Euler pole and the TFF varies from 400 to 500 km and the angle of rotation is about 25°, as deduced from paleomagnetic data. Thus, the post-Cretaceous displacement along the TFF is close to the total displacement and may be accounted for by rotation of the Fergana block, although this fault is Paleozoic in age.

The continuous continental crust certainly existed in Central Asia in Cenozoic times and no oceanic sutures of this age are known here. Therefore, any rotation should be reflected in the deformation of the crust. Structural trends along the northern border of the Fergana Basin are directed in a ENE–WSW direction (Fig. 8). The

magnitude of deformation (folds and thrusts) increases in a east-northeast direction. Recent uplifts here are usually triangular in form, have their bases coinciding with the TFF and tops pointing southwestwards. The largest of them is the Chatkal triangle, but there are others of smaller size (Fig. 12). The formation of the Chatkal triangle and other structural features of the Tien Shan has been connected with horizontal movements on geological grounds (Cobbold and Davy, 1988). With paleomagnetic data, it becomes evident that these triangles are compensation structures and the result of counter-clockwise rotation of the Fergana block, which created compressional strains along its northeastern and northwestern boundaries. In addition to triangular structures, Late Cenozoic thrusts and folds occur over a considerable part of the block, hence its internal deformation must be distributed in accordance with the observed fan of declinations.

The paleomagnetic directions are very different on both flanks of the Gubin Line thrust, which is usually regarded as the boundary between the Pamirs and Tien Shan domains (Fig. 12). The Cretaceous declination from locality 10, about 6 km north of the Gubin Line, is unrotated with respect to the Eurasian reference direction (Fig. 11), whereas Cretaceous declination (about

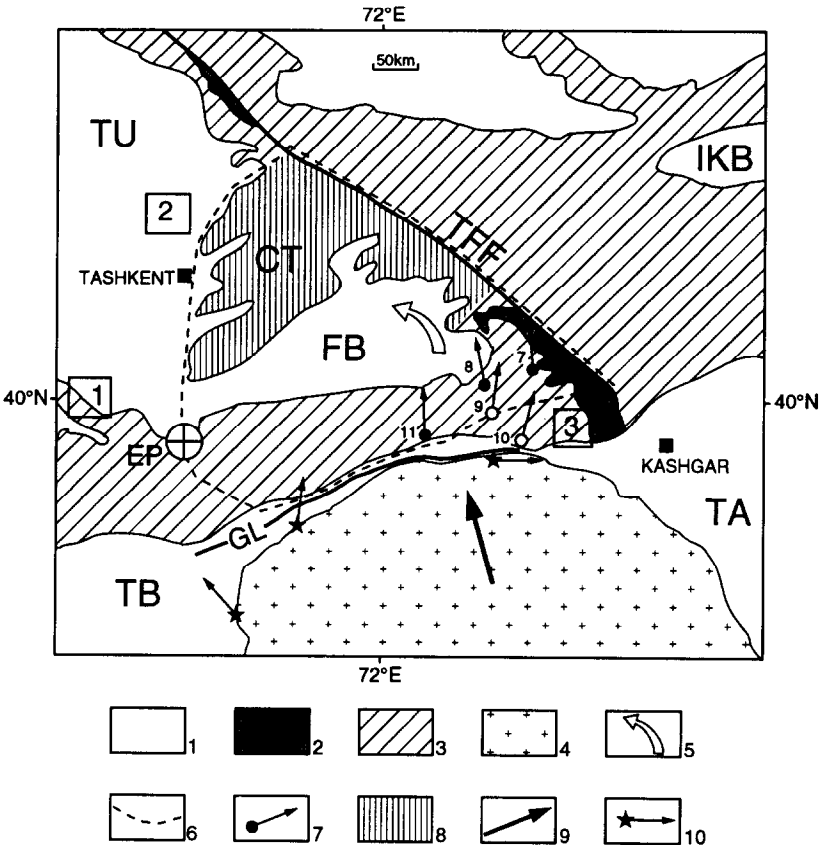



Fig. 12. Diagram showing the rotation of the Fergana block. *1* = Cenozoic cover; *2* = Jurassic basins joined to the TFF; *3* = the Tien Shan domain (mainly Paleozoic); *4* = the Pamirs domain; *5* = direction of the proposed rotation of the Fergana block; *6* = the boundary of the rotated block; *7* = some sampling localities in the Tien Shan numbered as in the text; dots = rotated, circles = unrotated, measured paleomagnetic directions shown as black arrows; *8* = the compensation structures accommodating rotation of the Fergana block, *CT* = Chatkal triangle; *9* = the direction of the Pamirs movement (Bazhenov and Burtman, 1986); *10* = Cretaceous sampling localities from the Pamirs External zone and paleomagnetic directions (Bazhenov and Burtman, 1986); numbers in rectangles denote the areas presumed unrotated with respect to Eurasia according to the proposed model (see text); *IKB* = Issyk-Kul basin; *TFF* = Talas-Fergana fault; *TU* = Turan plate; *FB* = Fergana basin; *TA* = Tarim platform; *GL* = Gubin Line thrust; *TD* = Tadjik depression; *EP* = Euler pole position, shown as circle with cross.

4 km to the south) is rotated about 70° clockwise (Bazhenov and Burtman, 1986). Thus, the southern boundary of the rotated block most probably coincides with this thrust.

The relative position of the Euler pole and the TFF (Fig. 12) suggests that the largest strike-slip displacement should be in the central part of the TFF, east of the Chatkal triangle. This is in agreement with structural observations. West of this compensation structure there should be no displacements along the TFF, which fits the geological data. Moving eastward along the fault, the strike-slip component decreases, whereas the compressive component increases. The Pamirs wedge most probably did not move due northward but was slightly deflected westward (Bazhenov and Burtman, 1986). Thus, the crust was compressed, without rotation, in the southeastern part of the area studied, thus accounting for the apparent lack of rotation here (Fig. 12). If this assumption is correct, there ought to be no rotation further eastward, between the China–Kirghizia border and Kashgar city in Tarim. The same reasoning accounts for the extinction of the TFF in the area where the narrow gap between the Tien Shan and Pamirs widens towards the Tarim basin.

## Conclusions

From paleomagnetic data, the rotation of the Fergana block is certainly post-Late Cretaceous in age. On the other hand, there are no angular unconformities from the Early Cretaceous up to the Miocene in this region, and the main folding took place even later, in the Late Miocene–Pliocene times. Therefore, the rotation and conjugated compensation structures are of the Late Cenozoic age too.

The proposed model accounts for the main structural features of the South Tien Shan and Fergana Basin. There are three areas where Cretaceous rocks outcrop, can be tested to verify or disprove this model (Fig. 12): (1) the western limb of the hypothetical orocline; (2) an area to the west of the Chatkal triangle; and (3) the northwestern part of Tarim (Fig. 12). According to the proposed model, there should be no Cenozoic counter-clockwise rotation, which can be checked by paleomagnetic studies. Moreover, since this the article was submitted, one of these predictions has already been tested: a good agreement between Cretaceous data and the reference direction has been reported from Tarim (Chen et al., 1992).

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